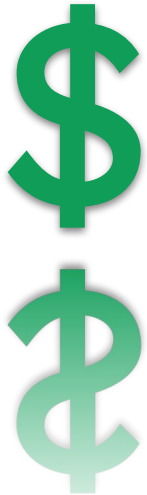
Abuse of AI in Finance

In-Depth Case Study By Funsho Orogun

Implementation of Ai in the Finance industry .



- Integration of transaction data, device fingerprints, IP logs
- Training ML models on historical fraud patterns
- Real-time scoring of every transaction
- Human-in-the-loop review for high-risk cases
- Continuous model retraining to adapt to new fraud tactics

Benefits .

- Higher fraud detection accuracy
- Real-time blocking of suspicious transactions
- Reduced financial losses
- Better customer protection
- Operational efficiency for fraud analysts

Challenges .

- False positives = customer frustration
- Model drift as criminals evolve tactics
- Data bias leading to unfair targeting
- Explainability issues with complex models
- Privacy concerns due to extensive monitoring
- Arms race between banks and fraudsters

Real world study : Hong Kong Deepfake Heist (2024) .

- Employee got invited to a video call with what appeared to be the CFO and his coworkers .
- Entire meeting was a **deepfake**, generated using AI voice and video models.
- Result: **$25 million stolen**

HK's first deepfake video conference scam involving HK$200 million

Ai tech used By the Culprit .

- Deepfake video & voice cloning
- Synthetic identities generated with AI
- Automated phishing using LLMs
- AI-powered social engineering scripts

Ai tech used to Catch the Culprit .

- Supervised ML models (gradient boosting, neural networks)
- Unsupervised anomaly detection (autoencoders, clustering)
- Graph-based models to detect fraud rings
- Behavioral biometrics (typing, device patterns)
- Real-time scoring engines for . instant . transaction risk evaluation

Ethical Considerations and Societal Impact.

Positive

- Increased trust in digital banking
- Reduced systemic financial risk
- Safer online transactions

Negative (Ethical Considerations)

- Normalization of financial surveillance
- Potential financial exclusion for “non-typical” users
- Escalation of AI-enabled crime
- Psychological harm from deepfake impersonation

Future Trends .

- Graph neural networks for complex fraud networks
- Multimodal AI combining voice, video, and behavior
- Federated learning for privacy-preserving fraud detection
- Regulatory frameworks (AI governance, transparency laws)
- Defensive generative AI to simulate new fraud scenarios

Recommendations .

- Strengthen AI governance and transparency
- Implement bias monitoring and fairness audits
- Improve employee training on deepfake awareness
- Use multi-factor verification for high-value transfers
- Invest in privacy-preserving AI
- Conduct regular red-team testing against AI-enabled attacks

Conclusion .

AI is both a powerful tool and a powerful threat in finance. Deepfake fraud shows how AI can be abused at scale. Ethical, transparent, and well-governed AI systems are essential to protect customers and maintain trust.